

BASEBALL AND STATISTICS

SESSION 4- DATA SCIENCE, AI, FAIR PLAY, ETHICS, AND OUR ADVICE TO TWINS MANAGMENT IN BUILDING THEIR ROSTER FOR THE FUTURE

Review of Session 3

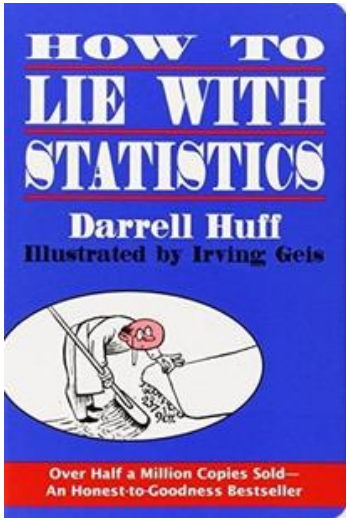
In-game metrics

- What the fans see
- What players and coaches see on the Ipad in the dugout
- What players and coaches use for pre-game planning
- PitchCom
- ABS

Kurt shared an Instagram site that provides a different take on how some of these metrics are computed.

Data Science and AI are statistics on steroids

Statistics can be used to prove past "truths" - the good old hot stove league



A 1954 Classic

Statistics can also be used to predict the future. This is our emphasis today.

- **How *data science* and *AI* come into play**
- **Effect on ethics and fair play as baseball moves ahead**
- **Completion of Tom R's Twins roster-analysis exercise from last week plus …**
- **A chance for you to advise Twins ownership on what they should do next**



DATA SCIENCE AND AI

FAIR PLAY AND “KEEPING THE CLUBHOUSE”

With all that predictive power that AI has, would teams use it in the dugout for real-time strategic decisions?

The Mets tried. According to whistle-blower and former Mets reliever Adam Ottavino, the Mets used a loophole in a customize tab on the Ipad in their dugout to run "*an AI program that was very expensive"* to help call pitches for PitchCom.

In a June 11, 2026 memo, MLB executive vice president Morgan Sword announced that the customize tab would be disabled. Some reactions from around the league:

- Aaron Judge: *Teams are making decisions off of AI? Man, that's just crazy*
- Blue Jays manager John Schneider termed it *a little weird that decisions could be swayed in real time*
- Diamondbacks manager Torey Lovullo*: AI is entering everyone's arena. You better get on it, or you're going to get rolled over by it*

LLM-based AI does not consider the physiology, neurochemistry, and emotions that make us human

Decisions ignoring that can lead to clubhouse tensions, as Red Sox manager Chad Tracy found out

- When pulled from the game after 70 pitches in a recent playoff loss to the Yankees, pitcher Sonny Gray said: *I would've liked to have been trusted in that moment. I think I'm the best pitcher in the American League going through the order the third time. I think that's a fact. I would have liked to live or die with me in that moment. I think you can probably ask the (other players), and they felt the same way.*
- His teammate Wilson Contreras: *We have to do a lot of different stuff and it's not enough when you have talent or you have data or you have computers telling you what to do. This is baseball and the data is here to help us. But at the end of the day, you're going to need to trust experience and veteran guys to win games. There's a lot more stuff that I would love to say but I'll stay like that*
- Having very recently been awarded a new three-year contract, Tracy might have a tough time re-capturing the clubhouse next spring.

LLM-based AI does not consider the physiology, neurochemistry, and emotions that make us human (cont.)

Going back to Reggie Jackson (and probably before), we've used phrases like "comes through in the clutch", "wired differently", and "slows things down" to describe those great players who seem to rise to the occasion better than others.

It's not far-fetched to imagine an AI of the future *that's not LLM-based* and will consider these factors. Its importance will be less about a manager's making statistically based decisions and more about how some athletes are better able to train their entire physiology into becoming "Reggie-Jackson-like".

Lucas Giolito is a very good MLB pitcher who has successfully battled some arm health problems. Even more significantly he learned how to use neurofeedback treatments to cope with anxiety that kept him from "slowing things down" in those pressure-packed moments that are so unique to baseball.



DATA SCIENCE AND AI

ETHICS

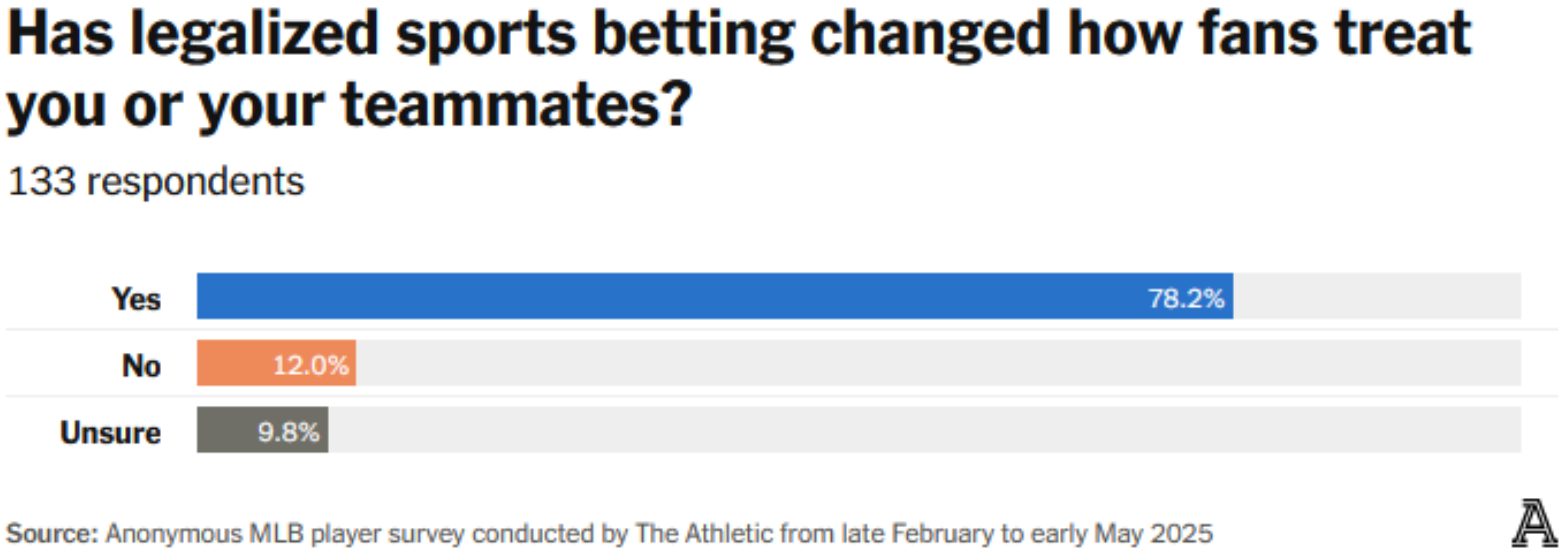
Data Science and AI in baseball's burgeoning alliance with online betting platforms

- MLB teams keep striking partnerships with prediction markets: What happens next?
- This was on full display in ads during recent World Series - DraftKings, FanDuel, BetMGM, Polymarket, Kalshi, Novig, …
- The real-time predictions you saw on TV during the series are not the odds you get when you go to place a bet. The odds the gambling platforms provide for their users are tempered by …

Pressures that the betting platforms place on players

- Rigged pitches, coded messages and competing defenses: A guide to the upcoming Clase-Ortiz trial
- Players are regularly targeted when they don't come through for a fan who's placed a bet on them: 'People suck, dude': MLB players express dismay at the impact of sports betting on fan interactions

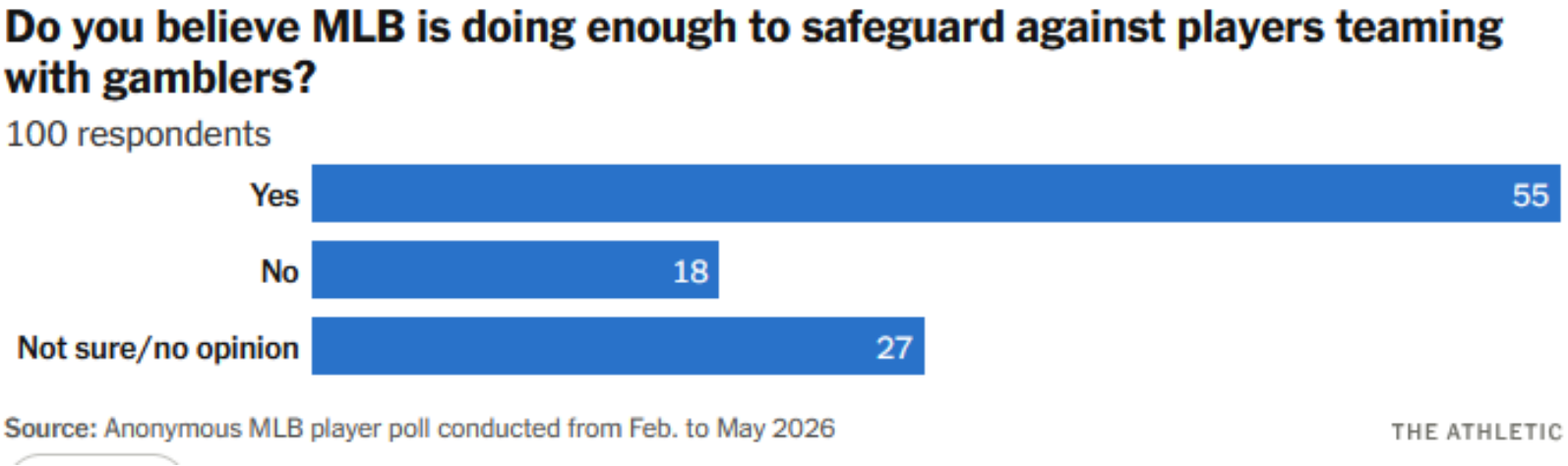
Results from The Athletic's (non-scientific) survey



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ROSTER ANALYSIS AND BUILDING

NEXT YEAR – TWINS VS BREWERS WORLD SERIES

Milwaukee Brewers

Roster Style: Young and athletic

Milwaukee began to put more emphasis on athleticism and contact hitting when it came to their position players, while encouraging situational awareness all through their system. This year's club is the most advanced manifestation of that approach.

On the pitching side, the Brewers are a mix of internally developed hurlers along with a number of acquired pitchers. Milwaukee has had an inordinate amount of success in coaxing better results from such pitchers.

Payroll Power: Bottom Third

The highest the Brewers have ranked in Opening Day payroll during their current 10-year run of success is 17th, back in 2019. They've won four straight National League Central titles while ranking no higher than 20th

Milwaukee Brewers

Homegrown Percentage: 32%

The Brewers' 40-man roster is 32% homegrown, ranking 20th; the 26-man roster is at 38%, the highest of the four remaining teams. The Brewers tend to acquire external players via trades and waiver claims as opposed to free agency.

Signature Signing: prospect extensions

The signings that have been in step with the Brewers' philosophy have been extensions, especially to some of their most talented prospects. Three starters (Jackson Chourio, Cooper Pratt and Luis Lara) each signed extensions prior to reaching the majors.

Best Trade They’ve Made: Christian Yelich

From Miami in January 2018 for four players. He won NL MVP honors in his first season and finished second in the balloting the season after. The four players the Brewers sent to Miami in the deal are all out of the majors and finished with a combined minus 5.9 career bWAR.

Milwaukee Brewers

Biggest Reclamation Project Success - Jake Bauers

This area is a specialty of the Brewers, so a typical season would see at least a couple of examples. Before 2026, Bauers had a career OPS+ of 84 and was below replacement value (minus-1.2 bWAR). This season, Bauers had a 140 OPS+, put up 3.3 bWAR and led the Brewers with 26 home runs.

Historical Baggage: Overpacked

This is an esoteric way to describe the fact that the Brewers are one of five franchises never to win a title.

Minnesota Twins
Roster Style:

Payroll Power:

The Twins ranked 24th in team payroll in 2026 (or 23 or 22, depending on source)

Minnesota Twins

Homegrown Percentage: 50%

Signature Signing:

Best Trade They've Made:

Pablo Lopez - Luis Arreaz

Mick Abel - Jhoan Duran

Taj Bradley - Griffin Jax

Joe Ryan - Nelson Cruz

Minnesota Twins

Biggest Reclamation Project Success:

Kody Clemens was DFA'd by Phillies; led 2026 Twins with 27 HRs

Historical Baggage:

The Twins last won the World Series in 1991

Longer droughts - Guardians, Padres, Brewers, Mariners, Pirates, Orioles, Tigers, Mets, Athletics, Reds

Twins lost 18 straight postseason games from 2004-2020

A Twins roster building exercise with a jump start from AI

- We asked Perplexity AI to assume the MLB negotiations lead to no significant changes and that Twins ownership is willing to add approximately $50M to its present $111M payroll. Under these assumptions, describe two roster building strategies to maximize the probability of winning the AL pennant over the next three years. Strategy A is *Dodger-lite*, that is, sign the best available free agents next year, sacrificing some of your younger players. Strategy B is *Tampa Bay with a Better Budget*, that is, build a three-year core by not investing heavily in next year's free agents.
- For those who really like getting in the weeds (like Tom N), our course website offers Perplexity's complete report for this scenario and one in which Twins management is only willing to pony up $20M.

Here's what it said based on the top free agents

Features	Strategy A	Strategy B
Headline additions	Sign #3 Teruki Sato (5/$180M); trade for Hunter Goodman (catcher from Colorado who hit 41 HR), give up Larnach, Ober, and Kremer; let Jeffers go for the comp pick rather than extend him; sign RPs #12 Bednar and unranked Romero; Joe Ryan and Pablo Lopez leave after one more year of control	Re-sign Jeffers; trade for Xavier Edwards (2B from Miami); sign #13 Morejon and unranked Romero; extend Jenkins, Keaschall, Culpepper, Rodriguez, Bradley
Headline departures	Jeffers, Larnach, Ober, Kremer, Ryan, Lopez; Tait, Quick, Mercedes in trade for Goodman	Larnach, Ober (for prospects); Mercedes + a pitcher in the Edwards trade
2027 projected wins	~83.5–85.5	~79.5–81.5
2027 pennant probability	~4.5-8%	~3-5%
3-year pennant probability	~18-23%	~17-21%
2027 payroll	~$159.4M	~$148.5M (~$13M held in reserve)
Prospect capital	Spent (in Goodman trade)	Largely retained; only Mercedes and a pitcher traded
Signature risk	Sato and his $180M guarantee; thin farm system	Good-not-great 2027; ownership impatience in waiting

The Athletic Top 40 MLB Free Agents for 2026-27

Rank	Name	Position	WAR (2026)
1	Tarik Skubal	LHP	4.6
2	Freddy Peralta	RHP	0.7
3	Teruaki Sato	3B	N/A
4	Michael King	RHP	4.3
5	Randy Arozarena	LF	5.7
6	Seiya Suzuki	RF	5.1
7	Jake Bauers	1B/OF	3.3
8	Jazz Chisholm Jr.	2B	1.7
9	Brandon Lowe	2B	5.4
10	Clay Holmes	RHP	2.0

Top 40 (11-20)

Rank	Name	Position	WAR (2026)
11	Adrian Morejon	LHP	1.9
12	David Bednar	RHP	1.9
13	Nick Martinez	RHP	3.9
14	Foster Griffin	LHP	3.3
15	Trevor Rogers	LHP	2.2
16	Casey Mize	RHP	1.3
17	Luis Arraez	2B	2.9
18	Kaima Taira	RHP	N/A
19	Ian Happ	LF	2.9
20	Robbie Ray	LHP	2.3

Top 40 (21-30)

Rank	Name	Position	WAR (2026)
21	Shota Imanaga	LHP	2.0
22	Raisel Iglesias	RHP	1.4
23	Gleyber Torres	2B	2.0
24	Alec Bohm	1B/3B	-0.2
25	Kevin Gausman	RHP	1.1
26	Shane Bieber	RHP	0.6
27	Hiromi Itoh	RHP	N/A
28	Trent Grisham	CF	0.6
29	Taylor Ward	LF	1.7
30	Daulton Varsho	CF	0.9

Top 40 (31-40)

Rank	Name	Position	WAR (2026)
31	Tyler Stephenson	C	2.5
32	Ryan Jeffers	C	2.8
33	Alex Vesia	LHP	1.5
34	Paul Blackburn	RHP	3.1
35	Mauricio Dubón	INF/OF	2.7
36	Joc Pederson	DH	1.7
37	Gregory Soto	LHP	1.2
38	J.P. Crawford	SS/3B	0.9
39	Jonathan Loáisiga	RHP	1.7
40	Sean Newcomb	LHP	2.0